

Email Generation Assistant — Final Report

This report documents the prompting approach, custom evaluation metrics, raw evaluation results, strategy comparison, and production recommendation.

1. Project Summary

A working Email Generation Assistant was built to produce professional emails from structured inputs: Intent, Key Facts, and Tone. Two prompting strategies were compared across 10 evaluation scenarios using three custom metrics.

2. Prompting Approach and Template

Prompting Approach: Structured Prompting with LangChain (Strategy B)

This project uses **LangChain ChatPromptTemplate** chains backed by **Groq** (`ChatGroq`).

Strategy B uses **structured prompting** with a role instruction. The LangChain template separates:

- **System role**: Professional business email writer
- **Task**: Generate one complete email
- **Input fields**: Intent, Key Facts, Tone (template variables)
- **Output requirements**: Subject, Greeting, Body, Closing
- **No-hallucination rule**: Use only provided facts; placeholders for names

The generation chain is: `ChatPromptTemplate | ChatGroq | StrOutputParser`

This improves reliability by giving the model explicit section boundaries and constraints, reducing missed facts and invented details compared to the basic prompt.

Strategy A (basic) uses a minimal LangChain template without section rules or hallucination guardrails.

Structured Prompt Template (Strategy B):

[SYSTEM]

You are a professional business email writer. Follow instructions exactly and use only provided information.

[HUMAN]

Task

Generate one complete email based on the structured input below.

Input

Intent: Follow up after a product demo

Key Facts (every fact must appear naturally in the email):

- The demo was held on Monday
- The client asked for pricing details
- The proposal should be shared by Friday
- The client is interested in the automation dashboard

Tone: Formal

Output Requirements

Return the email with exactly these sections:

1. Subject: (one clear subject line)
2. Greeting: (e.g., Hi [Recipient Name],)
3. Body: (clear paragraphs covering the intent and all key facts)

4. Closing: (professional sign-off with [Sender Name])

- ## Rules
- Use ONLY information from the Key Facts. Do not invent names, prices, dates, attachments, or commitments not provided.
 - Match the requested tone throughout.
 - Keep the email professional and concise.
 - Use placeholders [Recipient Name] and [Sender Name] instead of inventing names.

Output Format
Subject: <subject line>

<greeting>

<body paragraphs>

<closing>

3. Evaluation Dataset Summary

10 unique scenarios with Intent, Key Facts, Tone, and Human Reference Email. Each scenario was run through Strategy A (basic prompt) and Strategy B (structured prompt).

4. Custom Metric Definitions and Logic

Fact Recall

Definition: Measures whether the generated email includes all required key facts from the input scenario.

Logic: For each key fact, check if its meaningful content appears in the email (case-insensitive keyword overlap). Score = included_facts / total_facts. Missing facts are listed in the result.

Tone and Format Accuracy

Definition: Measures whether the email matches the requested tone and contains all four expected sections: subject, greeting, body, and closing.

Logic: Format score (0.5 max): 0.125 per present section (subject, greeting, body, closing). Tone score (0.5 max): keyword/pattern match against tone-specific indicators. Final score = format_score + tone_score (0.0–1.0).

Conciseness and Fluency

Definition: Measures readability, polish, and conciseness of the generated email.

Logic: Starts at 1.0 and applies penalties: excessive length (>180 words), very short body (<20 words), repeated sentences, grammar issues (double spaces, missing punctuation), or excessive wordiness patterns.

5. Raw Evaluation Results

Strategy	Fact Recall	Tone & Format	Conciseness	Overall
basic	1.00	0.98	0.96	0.98
structured	1.00	0.92	0.95	0.96

Sample per-scenario scores (first 5 entries):

[
{

```

"scenario_id": "SCN-001",
"strategy": "basic",
"scores": {
  "fact_recall": 1.0,
  "tone_and_format_accuracy": 1.0,
  "conciseness_and_fluency": 1.0
},
"overall_average": 1.0,
"details": {
  "fact_recall": {
    "included_count": 4,
    "total_count": 4,
    "included": [
      "The demo was held on Monday",
      "The client asked for pricing details",
      "The proposal should be shared by Friday",
      "The client is interested in the automation dashboard"
    ],
    "missing": [],
    "summary": "4 of 4 key facts included"
  },
  "tone_and_format_accuracy": {
    "sections_present": {
      "subject": true,
      "greeting": true,
      "body": true,
      "closing": true
    },
    "missing_sections": [],
    "tone_requested": "Formal",
    "tone_indicator_hits": 3,
    "format_score": 0.5,
    "tone_score": 0.5
  },
  "conciseness_and_fluency": {
    "word_count": 157,
    "sentence_count": 7,
    "penalties": []
  }
},
{
  "scenario_id": "SCN-001",
  "strategy": "structured",
  "scores": {
    "fact_recall": 1.0,
    "tone_and_format_accuracy": 1.0,
    "conciseness_and_fluency": 0.9
  },
  "overall_average": 0.9667,
  "details": {
    "fact_recall": {
      "included_count": 4,
      "total_count": 4,
      "included": [
        "The demo was held on Monday",

```

6. Strategy Comparison

Recommended strategy: **basic**
Reason: Highest overall average (0.98 vs 0.96) and better scores across the three custom metrics.
Overall averages — basic: 0.98, structured: 0.96

7. Failure Mode Analysis

Biggest failure mode of structured: Lowest average on Tone and Format Accuracy (0.92): inconsistent formatting and tone misalignment.

8. Final Recommendation

Use basic — more complete, tone-accurate, and polished emails across the same 10 scenarios.